

- (i) Give the molecular formula of oct-1-en-3-ol. [1]

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- (ii) State a reagent that can be used for Stage 2 and name the type of reaction mechanism taking place. [2]

Reagent(s)

Type of reaction mechanism

- (iii) State a reagent that can be used for Stage 3. [1]

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- (b) An isomer of oct-1-en-3-ol was reacted with alkaline iodine and produced a yellow precipitate.

- (i) Give the formula of the yellow precipitate. [1]

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- (ii) Suggest a structural formula for this isomer and identify the group that enables it to produce this yellow precipitate. [2]

Structural formula

Group

- (c) Octane-2,3-dione is a pale yellow liquid and octane-2,3-diol is a colourless material.

- (i) State a reagent that can be used to produce octane-2,3-diol from octane-2,3-dione. [1]

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- (ii) Suggest how the rate of the reaction in part (i) could be studied. [1]

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